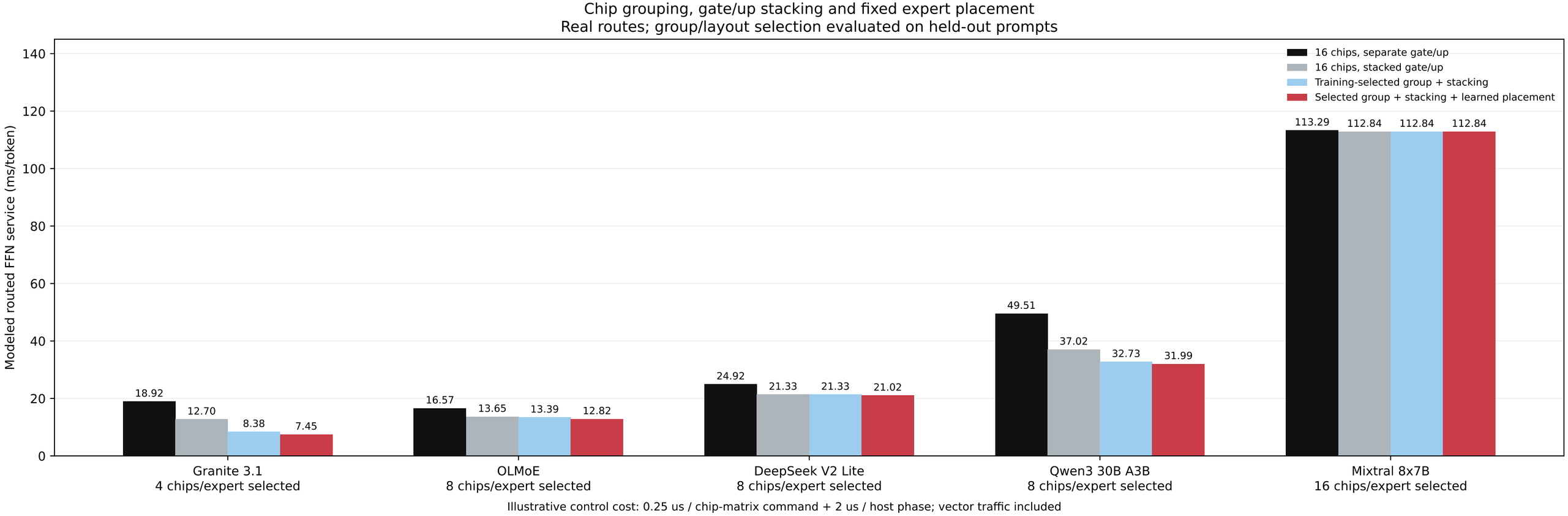
Chip grouping, gate/up stacking and fixed expert placement
Real routes; group/layout selection evaluated on held-out prompts
Modeled routed FFN service (ms/token)
0
20
40
60
80
100
120
140
16 chips, separate gate/up
16 chips, stacked gate/up
Training-selected group + stacking
Selected group + stacking + learned placement
18.92
12.70
8.38
7.45
16.57
13.65
13.39
12.82
24.92
21.33
21.33
21.02
49.51
37.02
32.73
31.99
113.29
112.84
112.84
112.84
Granite 3.1
4 chips/expert selected
OLMoE
8 chips/expert selected
DeepSeek V2 Lite
8 chips/expert selected
Qwen3 30B A3B
8 chips/expert selected
Mixtral 8x7B
16 chips/expert selected
Illustrative control cost: 0.25 us / chip-matrix command + 2 us / host phase; vector traffic included